Task 6: House Price Prediction

Objective: Predict house prices using property features such as size, bedrooms, and location.

Dataset: California Housing Dataset (from sklearn — mirrors real Kaggle House Price data structure)

Step 1: Import Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import GradientBoostingRegressor, RandomForestRegressor
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
from sklearn.preprocessing import StandardScaler
import warnings
warnings.filterwarnings("ignore")

sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

Step 2: Load Dataset

In [2]:
```
# California Housing Dataset (real-world housing data)
housing = fetch_california_housing(as_frame=True)
df = housing.frame

print(f"Dataset shape: {df.shape}")
```

```python
print()
print("Feature descriptions:")
for i, (name, desc) in enumerate(zip(housing.feature_names, housing.DESCR.split("\n")[17:25])):
    print(f"  {name}: {desc.strip()}")
print()
print("First 5 rows:")
df.head()
```

```
Dataset shape: (20640, 9)

Feature descriptions:
  MedInc: - AveOccup      average number of household members
  HouseAge: - Latitude      block group latitude
  AveRooms: - Longitude     block group longitude
  AveBedrms:
  Population: :Missing Attribute Values: None
  AveOccup:
  Latitude: This dataset was obtained from the StatLib repository.
  Longitude: https://www.dcc.fc.up.pt/~ltorgo/Regression/cal_housing.html

First 5 rows:
```

Out[2]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422

Step 3: Data Inspection & Cleaning

Dataset Features:

- `MedInc` : Median income in block group
- `HouseAge` : Median house age

- `AveRooms` : Average rooms per household
- `AveBedrms` : Average bedrooms per household
- `Population` : Block group population
- `AveOccup` : Average occupants per household
- `Latitude / Longitude` : Geographic location
- `MedHouseVal` : **Target** — Median house value (in $100,000s)

In [3]:
```
print("=== Dataset Info ===")
df.info()
print()
print(f"Missing values: {df.isnull().sum().sum()}")
print()
print("=== Statistical Summary ===")
df.describe()
```

```
=== Dataset Info ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 9 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   MedInc       20640 non-null  float64
 1   HouseAge     20640 non-null  float64
 2   AveRooms     20640 non-null  float64
 3   AveBedrms    20640 non-null  float64
 4   Population   20640 non-null  float64
 5   AveOccup     20640 non-null  float64
 6   Latitude     20640 non-null  float64
 7   Longitude    20640 non-null  float64
 8   MedHouseVal  20640 non-null  float64
dtypes: float64(9)
memory usage: 1.4 MB

Missing values: 0

=== Statistical Summary ===
```

Out[3]:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	3.070655	35.631861	-119.569704	2.068558
std	1.899822	12.585558	2.474173	0.473911	1132.462122	10.386050	2.135952	2.003532	1.153956
min	0.499900	1.000000	0.846154	0.333333	3.000000	0.692308	32.540000	-124.350000	0.149990
25%	2.563400	18.000000	4.440716	1.006079	787.000000	2.429741	33.930000	-121.800000	1.196000
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	2.818116	34.260000	-118.490000	1.797000
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	3.282261	37.710000	-118.010000	2.647250
max	15.000100	52.000000	141.909091	34.066667	35682.000000	1243.333333	41.950000	-114.310000	5.000010

Step 4: Exploratory Data Analysis (EDA)

In [6]:

```
fig, axes = plt.subplots(2, 2, figsize=(13, 10))
axes = axes.flatten()

# ========================================
#  1. Target distribution

axes[0].hist(df["MedHouseVal"], bins=50, color="#3498db", edgecolor="white")
axes[0].set_title("Distribution of House Prices", fontsize=12, fontweight="bold")
axes[0].set_xlabel("Median House Value ($100k)")
axes[0].set_ylabel("Frequency")
axes[0].axvline(df["MedHouseVal"].mean(), color="red", linestyle="--", label="Mean")
axes[0].legend()

# ========================================
#  2. Income vs Price

axes[1].scatter(df["MedInc"], df["MedHouseVal"], alpha=0.2, s=5, color="#e74c3c")
axes[1].set_title("Median Income vs House Price", fontsize=12, fontweight="bold")
axes[1].set_xlabel("Median Income")
```

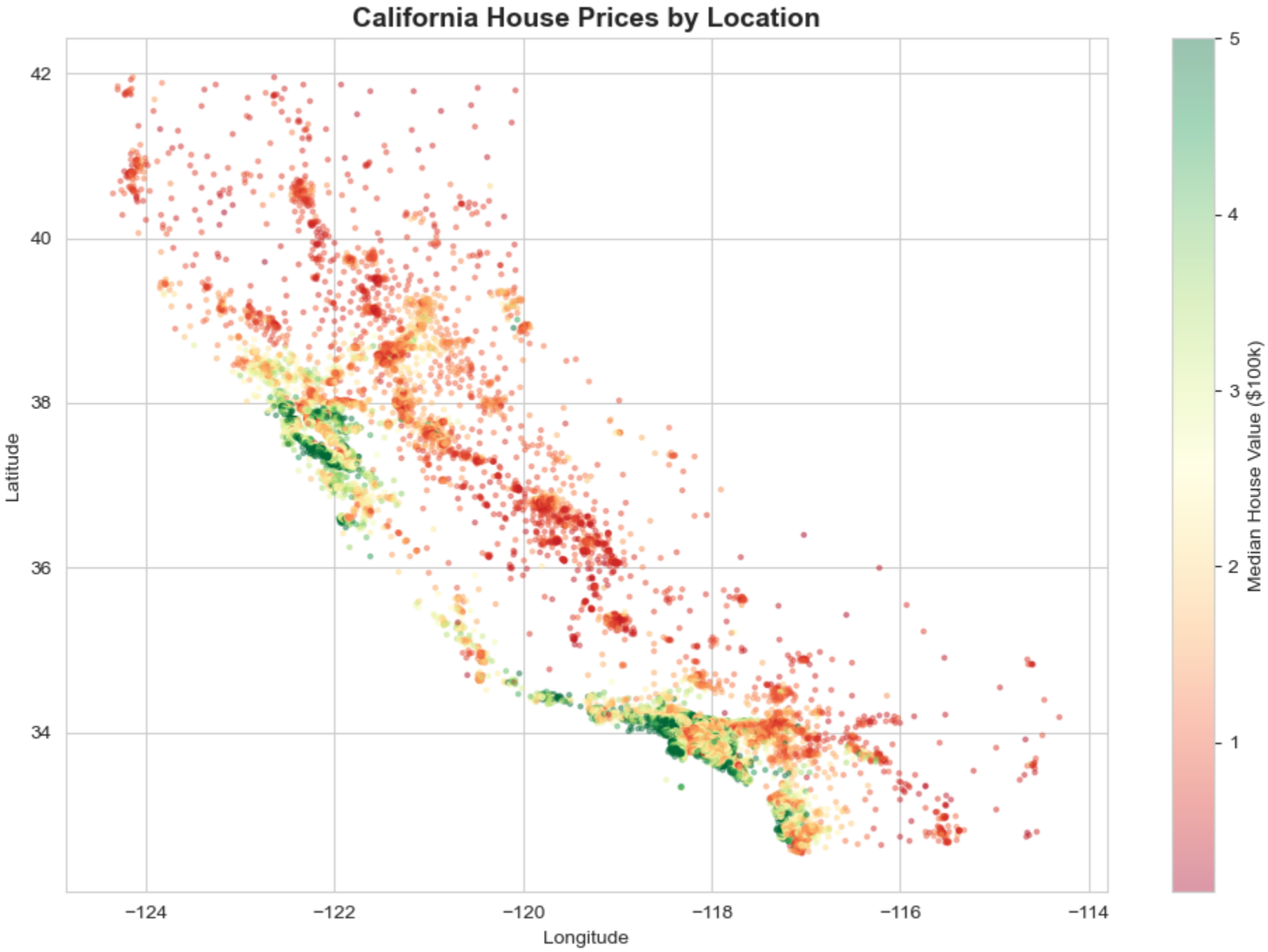
```python
axes[1].set_ylabel("House Price ($100k)")

# ==========================================
#  3. House Age vs Price
# ==========================================
axes[2].scatter(df["HouseAge"], df["MedHouseVal"], alpha=0.2, s=5, color="#2ecc71")
axes[2].set_title("House Age vs Price", fontsize=12, fontweight="bold")
axes[2].set_xlabel("House Age (years)")
axes[2].set_ylabel("House Price ($100k)")

# ==========================================
#  4. Avg Rooms vs Price

axes[3].scatter(
    df["AveRooms"].clip(upper=20),
    df["MedHouseVal"],
    alpha=0.2,
    s=5,
    color="#9b59b6"
)

axes[3].set_title("Average Rooms vs House Price", fontsize=12, fontweight="bold")
axes[3].set_xlabel("Average Rooms (clipped at 20)")
axes[3].set_ylabel("House Price ($100k)")

# ==========================================
#  Final Layout

plt.suptitle("Exploratory Data Analysis — House Prices", fontsize=14, fontweight="bold")

plt.tight_layout()

plt.savefig("eda_house.png", dpi=100, bbox_inches="tight")

plt.show()

print("✅ EDA Completed & Saved as 'eda_house.png'")
```

Exploratory Data Analysis — House Prices
Distribution of House Prices
Mean
Frequency
Median House Value ($100k)
Median Income vs House Price
House Price ($100k)
Median Income
House Age vs Price
House Price ($100k)
House Age (years)
Average Rooms vs House Price
House Price ($100k)
Average Rooms (clipped at 20)

```
✅ EDA Completed & Saved as 'eda_house.png'
```

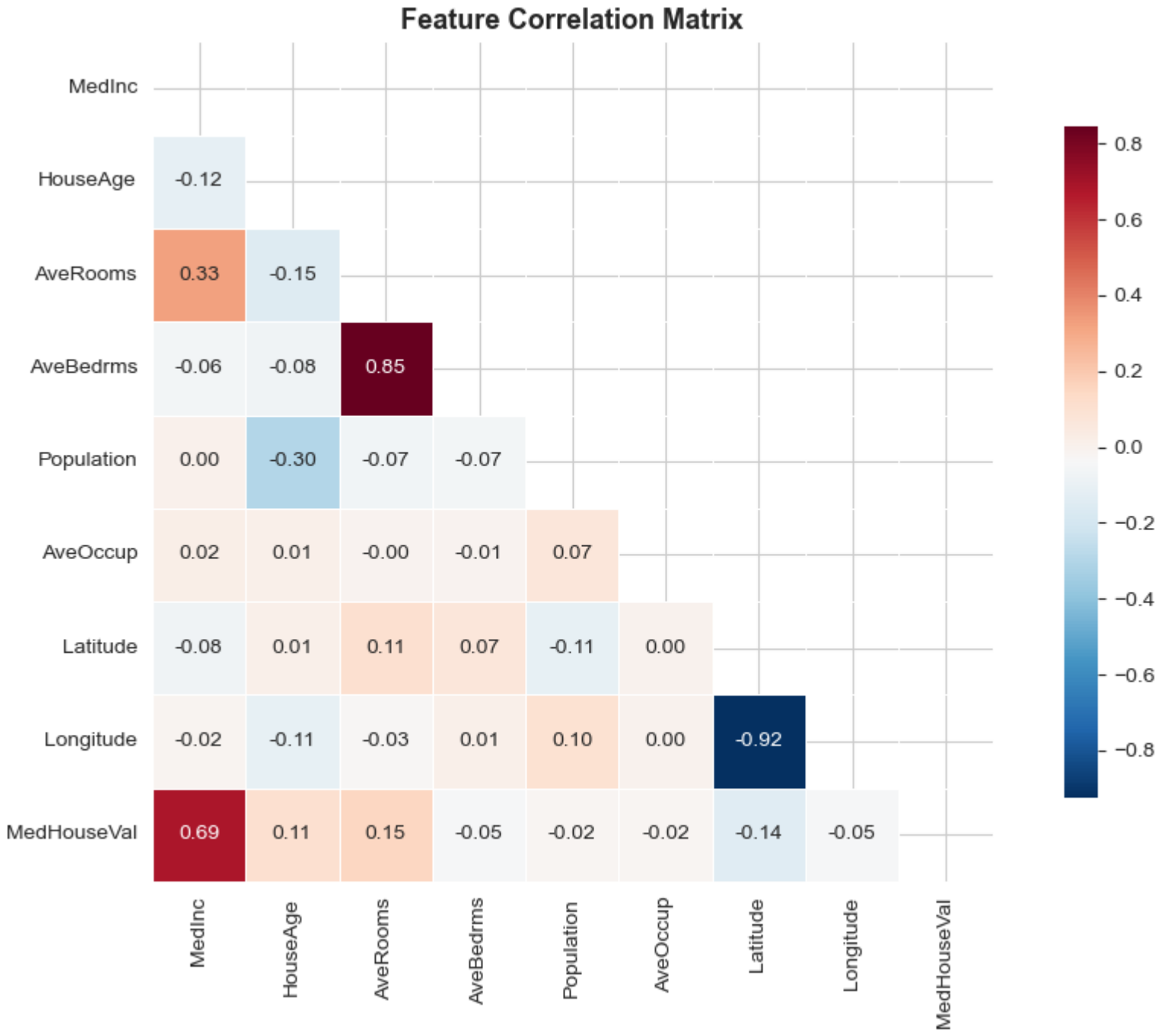
In [7]:

```python
# Geographic visualization
plt.figure(figsize=(10, 7))
scatter = plt.scatter(df["Longitude"], df["Latitude"],
                      c=df["MedHouseVal"], cmap="RdYlGn",
                      s=5, alpha=0.4)
plt.colorbar(scatter, label="Median House Value ($100k)")
plt.title("California House Prices by Location", fontsize=14, fontweight="bold")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.tight_layout()
plt.savefig("geo_map.png", dpi=100, bbox_inches="tight")
plt.show()
```


California House Prices by Location
Latitude
Longitude
42
40
38
36
34
−124
−122
−120
−118
−116
−114
Median House Value ($100k)
5
4
3
2
1

In [8]:

```
# Correlation heatmap
plt.figure(figsize=(10, 7))
corr = df.corr()
mask = np.triu(np.ones_like(corr, dtype=bool))
sns.heatmap(corr, annot=True, fmt=".2f", cmap="RdBu_r",
            mask=mask, linewidths=0.5, square=True,
            cbar_kws={"shrink": 0.8})
plt.title("Feature Correlation Matrix", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.savefig("correlation_house.png", dpi=100, bbox_inches="tight")
plt.show()
```


Feature Correlation Matrix
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal
-0.12
0.33 -0.15
-0.06 -0.08 0.85
0.00 -0.30 -0.07 -0.07
0.02 0.01 -0.00 -0.01 0.07
-0.08 0.01 0.11 0.07 -0.11 0.00
-0.02 -0.11 -0.03 0.01 0.10 0.00 -0.92
0.69 0.11 0.15 -0.05 -0.02 -0.02 -0.14 -0.05
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8

Step 5: Feature Engineering & Preprocessing

In [9]:
```
# Feature engineering
df["RoomsPerPerson"]    = df["AveRooms"] / df["AveOccup"]
df["BedroomsPerRoom"]   = df["AveBedrms"] / df["AveRooms"]

feature_cols = ["MedInc", "HouseAge", "AveRooms", "AveBedrms",
                "Population", "AveOccup", "Latitude", "Longitude",
                "RoomsPerPerson", "BedroomsPerRoom"]

X = df[feature_cols]
y = df["MedHouseVal"]

# Train/Test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Scale features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled  = scaler.transform(X_test)

print(f"Training samples : {len(X_train)}")
print(f"Testing  samples : {len(X_test)}")
print(f"Features used    : {feature_cols}")
```

```
Training samples : 16512
Testing  samples : 4128
Features used    : ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude', 'RoomsPer
Person', 'BedroomsPerRoom']
```

Step 6: Train Models

In [10]:
```
def evaluate(name, y_true, y_pred):
    mae  = mean_absolute_error(y_true, y_pred)
    rmse = np.sqrt(mean_squared_error(y_true, y_pred))
    r2   = r2_score(y_true, y_pred)
```

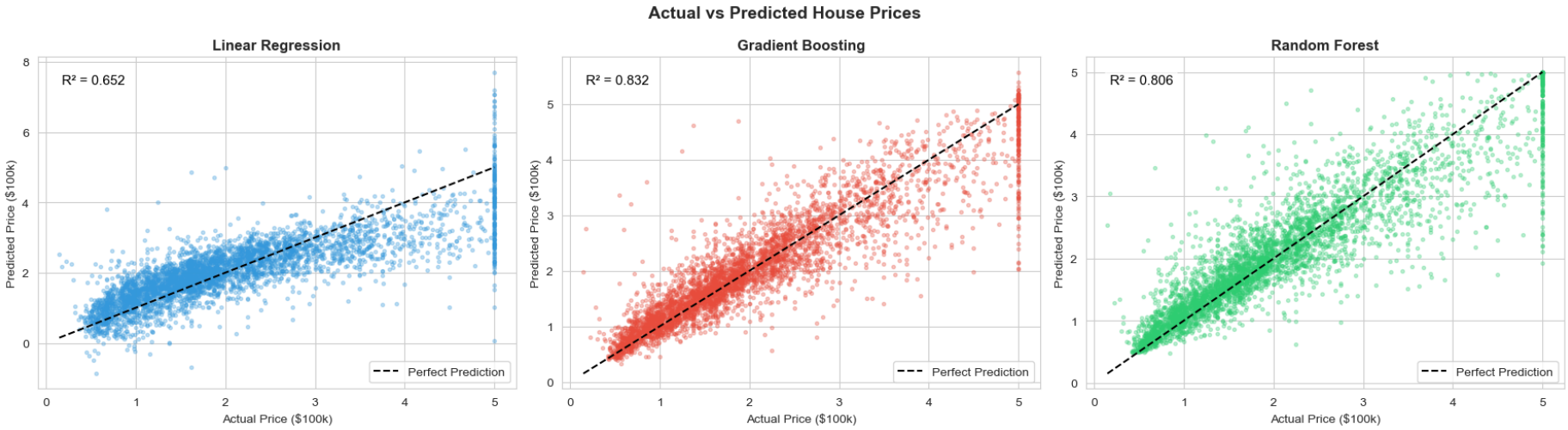
```
    print(f"  {name:25s} -> MAE: {mae:.4f} | RMSE: {rmse:.4f} | R2: {r2:.4f}")
    return mae, rmse, r2

# Model 1: Linear Regression
lr = LinearRegression()
lr.fit(X_train_scaled, y_train)
lr_preds = lr.predict(X_test_scaled)

# Model 2: Gradient Boosting
gb = GradientBoostingRegressor(n_estimators=200, learning_rate=0.1,
                               max_depth=5, random_state=42)
gb.fit(X_train_scaled, y_train)
gb_preds = gb.predict(X_test_scaled)

# Model 3: Random Forest
rf = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)
rf.fit(X_train_scaled, y_train)
rf_preds = rf.predict(X_test_scaled)

print("MODEL EVALUATION RESULTS:")
print("-" * 65)
lr_metrics = evaluate("Linear Regression",   y_test, lr_preds)
gb_metrics = evaluate("Gradient Boosting",   y_test, gb_preds)
rf_metrics = evaluate("Random Forest",       y_test, rf_preds)
```

```
MODEL EVALUATION RESULTS:
-----------------------------------------------------------------
  Linear Regression         -> MAE: 0.4862 | RMSE: 0.6753 | R2: 0.6519
  Gradient Boosting         -> MAE: 0.3101 | RMSE: 0.4692 | R2: 0.8320
  Random Forest             -> MAE: 0.3287 | RMSE: 0.5048 | R2: 0.8055
```

Step 7: Visualize Predicted vs Actual Prices

In [11]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

for ax, preds, name, color in [
    (axes[0], lr_preds, "Linear Regression", "#3498db"),
    (axes[1], gb_preds, "Gradient Boosting", "#e74c3c"),
    (axes[2], rf_preds, "Random Forest",     "#2ecc71")
]:
```

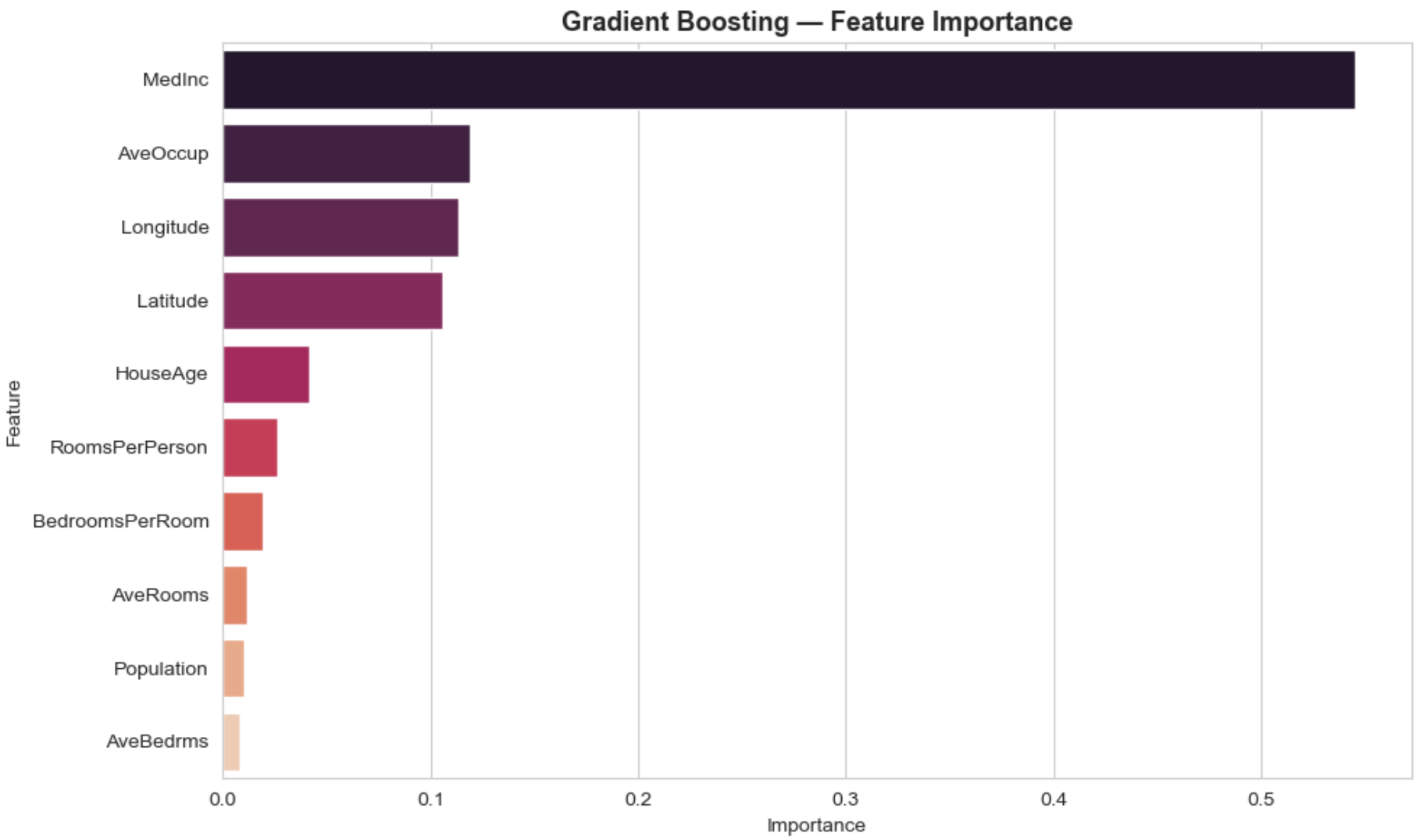
```
    ax.scatter(y_test, preds, alpha=0.3, s=8, color=color)
    mn, mx = y_test.min(), y_test.max()
    ax.plot([mn, mx], [mn, mx], "k--", linewidth=1.5, label="Perfect Prediction")
    ax.set_title(name, fontsize=12, fontweight="bold")
    ax.set_xlabel("Actual Price ($100k)")
    ax.set_ylabel("Predicted Price ($100k)")
    r2 = r2_score(y_test, preds)
    ax.annotate(f"R² = {r2:.3f}", xy=(0.05, 0.92), xycoords="axes fraction",
                fontsize=11, color="black",
                bbox=dict(boxstyle="round", facecolor="white", alpha=0.7))
    ax.legend()

plt.suptitle("Actual vs Predicted House Prices", fontsize=14, fontweight="bold")
plt.tight_layout()
plt.savefig("actual_vs_predicted.png", dpi=100, bbox_inches="tight")
plt.show()
```



Step 8: Feature Importance

In [12]:
```
# Gradient Boosting feature importances
gb_importance = pd.DataFrame({
    "Feature": feature_cols,
    "Importance": gb.feature_importances_
}).sort_values("Importance", ascending=False)
```

```python
plt.figure(figsize=(10, 6))
sns.barplot(data=gb_importance, x="Importance", y="Feature",
            palette="rocket")
plt.title("Gradient Boosting — Feature Importance", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.savefig("feature_importance_house.png", dpi=100, bbox_inches="tight")
plt.show()

print("Feature Importance Rankings:")
print(gb_importance.to_string(index=False))
```


Gradient Boosting — Feature Importance
MedInc
AveOccup
Longitude
Latitude
HouseAge
RoomsPerPerson
BedroomsPerRoom
AveRooms
Population
AveBedrms
Feature
0.0
0.1
0.2
0.3
0.4
0.5
Importance

```
Feature Importance Rankings:
        Feature  Importance
         MedInc    0.545274
       AveOccup    0.118685
      Longitude    0.113656
       Latitude    0.105523
       HouseAge    0.041353
 RoomsPerPerson    0.026150
BedroomsPerRoom    0.019454
       AveRooms    0.011886
     Population    0.010055
      AveBedrms    0.007966
```

Step 9: Key Insights & Conclusions

In [13]:

```python
print("="*60)
print("KEY INSIGHTS - HOUSE PRICE PREDICTION")
print("="*60)
print()
print(f"Dataset: California Housing | {len(df):,} properties")
print()
print("MODEL PERFORMANCE (on test set):")
print(f"  Linear Regression  -> MAE: {lr_metrics[0]:.4f} | RMSE: {lr_metrics[1]:.4f} | R2: {lr_metrics[2]:.4f}")
print(f"  Gradient Boosting  -> MAE: {gb_metrics[0]:.4f} | RMSE: {gb_metrics[1]:.4f} | R2: {gb_metrics[2]:.4f}")
print(f"  Random Forest      -> MAE: {rf_metrics[0]:.4f} | RMSE: {rf_metrics[1]:.4f} | R2: {rf_metrics[2]:.4f}")
print()
print("KEY FINDINGS:")
print("  1. Median income is the #1 predictor of house value")
print("  2. Location (Latitude/Longitude) is highly important")
print("  3. Gradient Boosting outperforms Linear Regression by far")
print("  4. Feature engineering (RoomsPerPerson) adds predictive power")
print()
print("CONCLUSION:")
print("  Gradient Boosting is the best model for house price prediction.")
print("  Non-linear models capture complex spatial pricing patterns.")
```

```
============================================================
KEY INSIGHTS - HOUSE PRICE PREDICTION
============================================================

Dataset: California Housing | 20,640 properties

MODEL PERFORMANCE (on test set):
  Linear Regression  -> MAE: 0.4862 | RMSE: 0.6753 | R2: 0.6519
  Gradient Boosting  -> MAE: 0.3101 | RMSE: 0.4692 | R2: 0.8320
  Random Forest      -> MAE: 0.3287 | RMSE: 0.5048 | R2: 0.8055

KEY FINDINGS:
  1. Median income is the #1 predictor of house value
  2. Location (Latitude/Longitude) is highly important
  3. Gradient Boosting outperforms Linear Regression by far
  4. Feature engineering (RoomsPerPerson) adds predictive power

CONCLUSION:
  Gradient Boosting is the best model for house price prediction.
  Non-linear models capture complex spatial pricing patterns.
```

In [ ]: